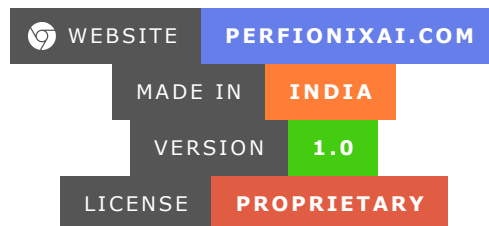




LAPLACIAN AI

India's First AI Workspace for Secure Enterprise Intelligence

Empowering Indian Enterprises with Sovereign AI Solutions



[Abstract](#) • [Introduction](#) • [Problem](#) • [Solution](#) • [Features](#) • [Tech Stack](#) • [Installation](#)

Company Information

Organization	Perfionix AI Technology Pvt Ltd
Product	Laplacian AI - Enterprise AI Workspace
Version	1.0
Industry	Artificial Intelligence / Enterprise Software

Abstract

Laplacian AI is India's pioneering AI Workspace platform, designed to address the critical need for data sovereignty and secure artificial intelligence solutions in the Indian enterprise ecosystem. As organizations increasingly rely on AI-powered tools for document analysis, data visualization, and

intelligent automation, concerns about data privacy, cross-border data transfer, and dependency on foreign AI infrastructure have become paramount.

Laplacian AI provides a comprehensive, locally-deployable AI workspace that combines **conversational AI**, **document intelligence (DocIQ)**, and **data visualization (VizIQ)** capabilities—all while ensuring that sensitive enterprise data never leaves Indian jurisdiction. Built on open-source foundation models and designed for on-premise deployment, Laplacian AI represents a strategic shift towards technological self-reliance in India's digital transformation journey.

Introduction

The AI Revolution and India's Opportunity

The global artificial intelligence market is projected to reach **\$1.8 trillion by 2030**, with enterprise AI applications driving significant productivity gains across industries. India, with its vast talent pool and rapidly digitizing economy, stands at a pivotal moment in this technological revolution.

However, the current landscape presents a critical challenge: **over 90% of enterprise AI tools used by Indian organizations** route sensitive data through servers located in the United States, European Union, and other foreign jurisdictions. This dependency creates risks related to:

- Data privacy and sovereignty
- Regulatory compliance (IT Act, DPDP Act 2023)
- Intellectual property protection
- Business continuity and geopolitical stability

Vision

"To establish India as a global leader in sovereign AI infrastructure, enabling enterprises to harness artificial intelligence without compromising data security or national interests."

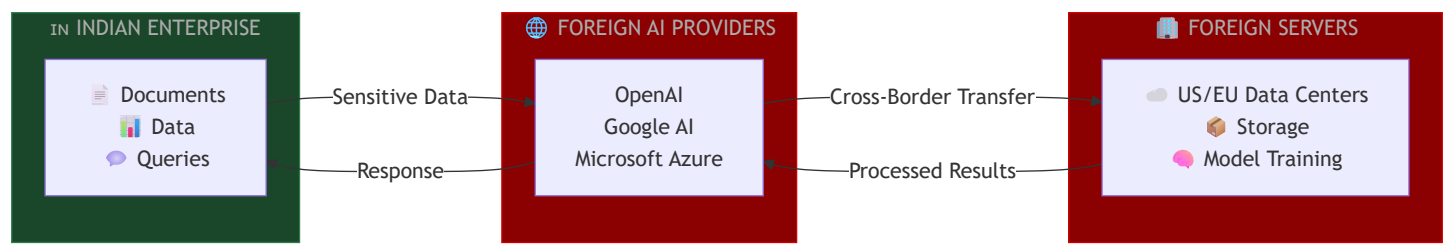
Mission

Laplacian AI is committed to:

1. **Data Sovereignty** - Ensuring Indian data remains within Indian borders
2. **Enterprise Empowerment** - Providing world-class AI tools accessible to all organizations
3. **Technological Independence** - Reducing dependency on foreign AI infrastructure

Problem Statement

The Data Sovereignty Crisis in Indian Enterprises



RISK: Sensitive Indian enterprise data processed and stored on foreign soil

Key Challenges Facing Indian Enterprises

Challenge	Impact	Scale
Data Privacy Violations	Sensitive corporate data exposed to foreign entities	78% of Indian enterprises use foreign AI tools
Regulatory Non-Compliance	Violation of IT Act 2000, DPDP Act 2023	₹250 Cr penalty provisions under DPDP
Intellectual Property Risk	Training data potentially used by foreign AI	Affects 65% of R&D-intensive organizations
Latency Issues	Degraded performance due to cross-border routing	200-500ms additional latency
Cost Escalation	Dollar-denominated pricing affecting budgets	15-30% YoY cost increase
Service Continuity	Dependency on foreign policy decisions	Single point of failure

Regulatory Landscape

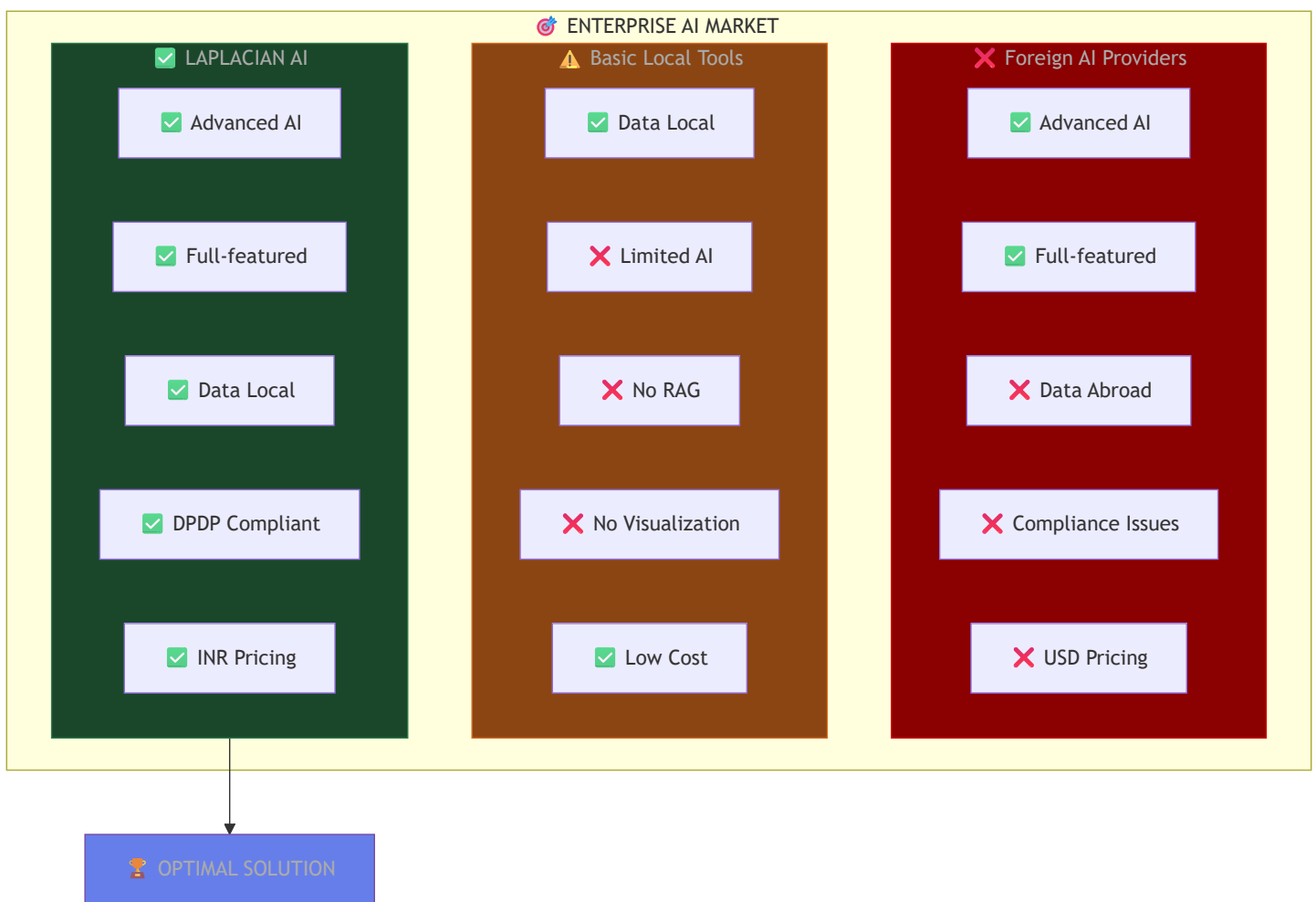
Digital Personal Data Protection Act, 2023 (DPDP)

- Mandates explicit consent for data processing
- Restricts cross-border data transfer to notified countries
- Imposes penalties up to ₹250 crore for violations
- Requires Data Protection Impact Assessments

IT Act 2000 & Amendments

- Section 43A: Compensation for failure to protect data
- Section 72A: Punishment for disclosure of information
- Intermediary guidelines requiring data localization

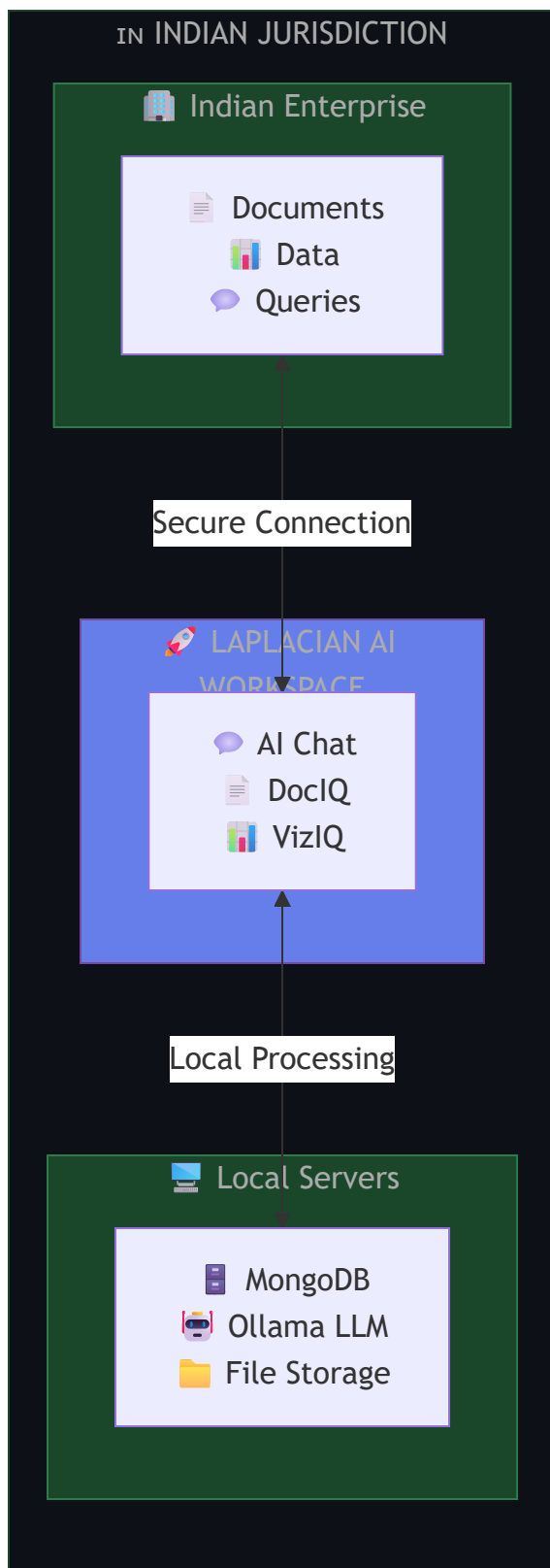
Market Gap Analysis



Solution

Laplacian AI: The Sovereign AI Workspace

Laplacian AI addresses these challenges through a comprehensive, locally-deployable AI workspace that delivers enterprise-grade capabilities while ensuring complete data sovereignty.



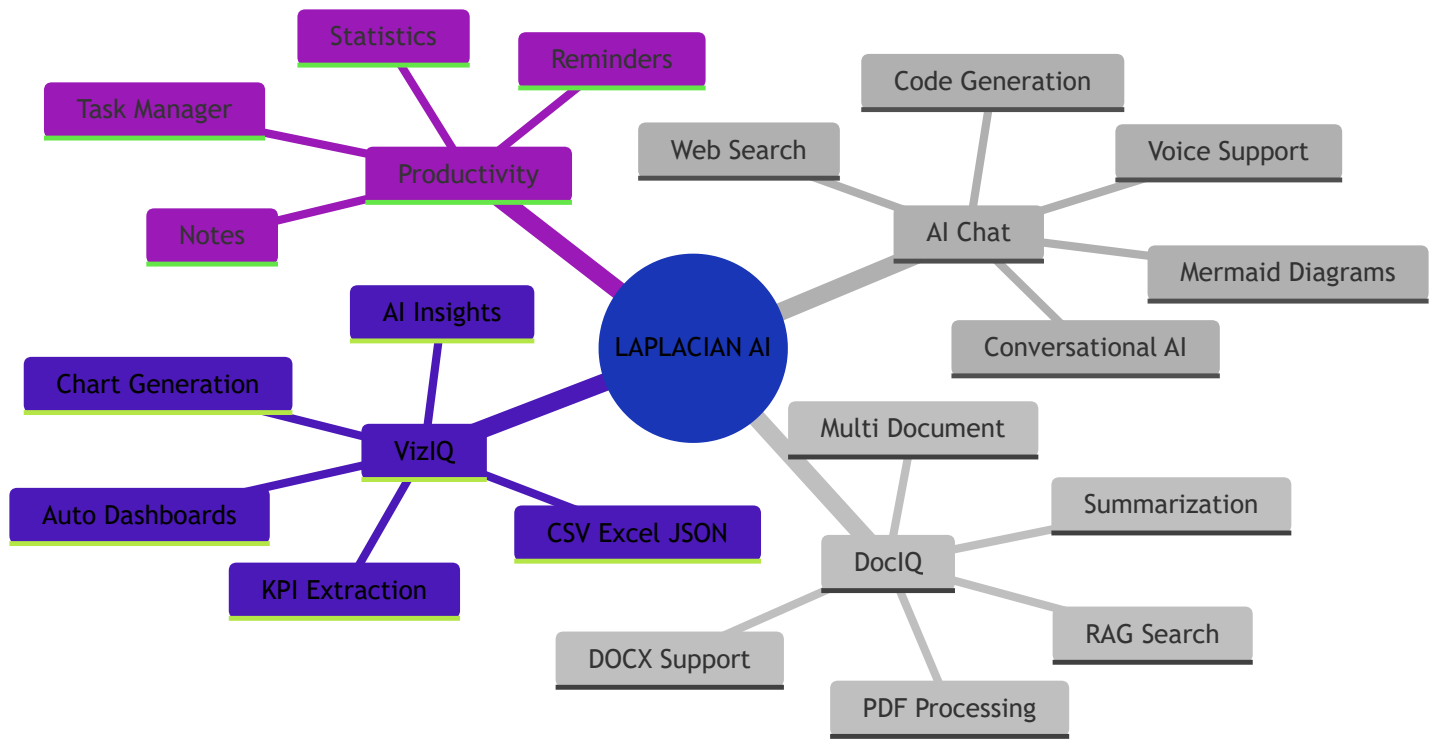
✅ **SECURE:** All data processed and stored within Indian jurisdiction

Core Value Propositions

Proposition	Description	Benefit
100% Data Sovereignty	All processing occurs on local/Indian infrastructure	Full DPDP compliance
Enterprise-Grade AI	Advanced LLM capabilities via Ollama	No compromise on quality
Unified Workspace	Chat + Documents + Data in one platform	Improved productivity
On-Premise Deployment	Deploy on your own servers	Complete control
Cost Efficiency	INR-based, predictable pricing	40-60% cost reduction
Customizable Models	Choose and fine-tune your AI models	Domain-specific excellence

Features

Laplacian AI provides a comprehensive AI workspace with three core intelligence modules and productivity tools:



AI Chat - Conversational Intelligence

The heart of Laplacian AI, providing intelligent conversational capabilities powered by locally-deployed language models.

Capabilities:

- Multi-model support (Core, Lite, Coder variants)
- Real-time web search integration
- Code generation with syntax highlighting
- Mermaid diagram visualization
- Architecture and flowchart generation
- Voice input/output support (ElevenLabs integration)
- Conversation history with edit/regenerate
- Context-aware responses

Use Cases:

- Code development and debugging
- Technical documentation
- Problem-solving and brainstorming
- Learning and explanation
- Architecture design

DocIQ - Document Intelligence

An advanced Retrieval-Augmented Generation (RAG) system for intelligent document analysis and question-answering.

Capabilities:

- Multi-format support (PDF, DOCX, DOC, TXT)
- Intelligent text chunking for optimal context
- Semantic search across documents
- Structured response formatting
- Automatic diagram generation
- Document summarization
- Cross-document analysis

Supported Formats:

Format	Extension	Processing
PDF Documents	.pdf	PyPDF2/pdfplumber extraction
Word Documents	.docx, .doc	python-docx parsing
Text Files	.txt	Direct text processing

Use Cases:

- Contract analysis and review
- Research paper summarization
- Policy document Q&A
- Technical manual navigation
- Legal document analysis

VizIQ - Data Intelligence

AI-powered data visualization and analytics platform that transforms raw data into actionable insights.

Capabilities:

- Auto-generated dashboards
- KPI extraction and display
- Multiple chart types (Bar, Line, Doughnut, Trend)
- Statistical analysis (Mean, Median, Min, Max)
- Data quality alerts
- AI-generated insights
- Column type detection

Supported Formats:

Format	Extension	Features
CSV	.csv	Full parsing, type detection
Excel	.xlsx, .xls	Multi-sheet support
JSON	.json	Nested structure handling

Generated Visualizations:

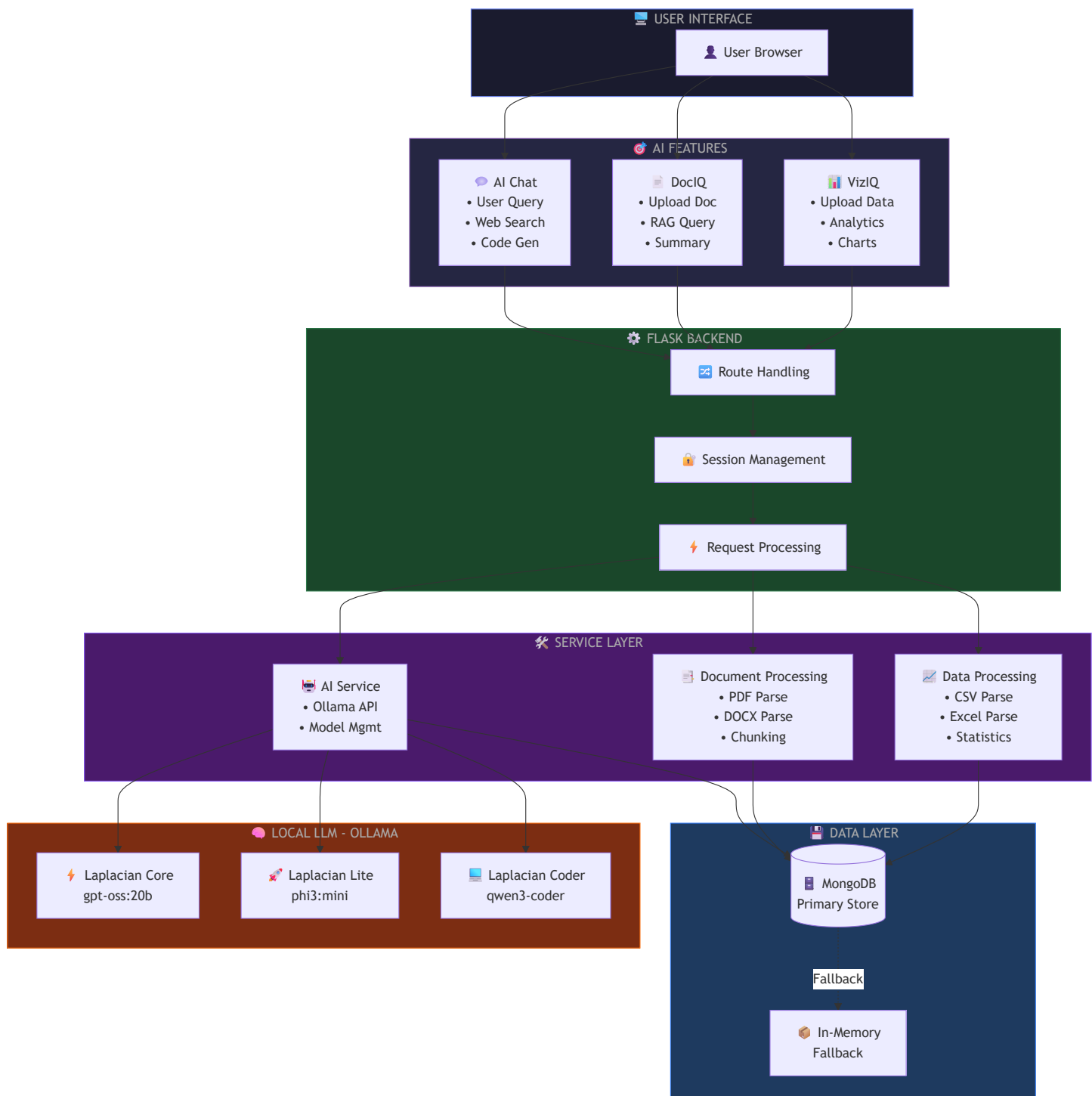
- Column charts (categorical aggregation)
- Distribution charts (pie/doughnut)
- Comparison charts (multi-metric)
- Trend analysis (time-series)

 Productivity Suite

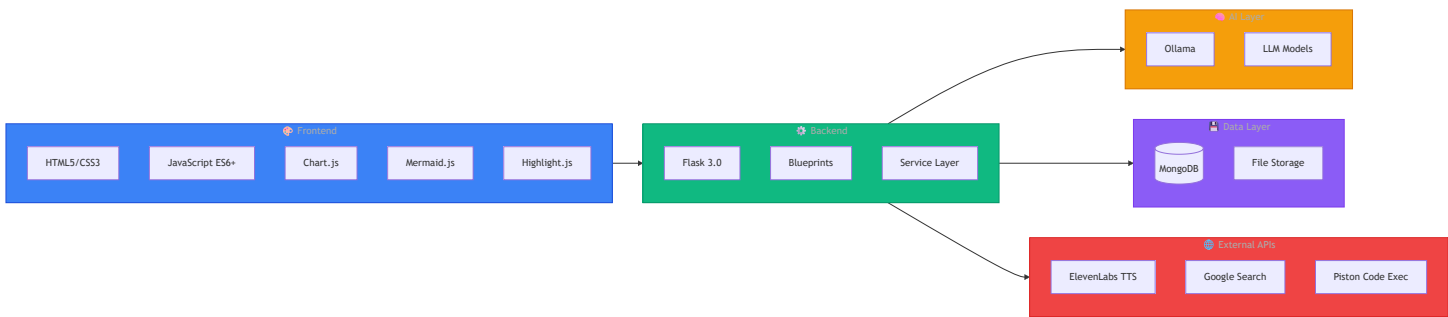
Integrated productivity tools for comprehensive workspace management:

Tool	Description	Features
Tasks	Task management with priorities	Create, complete, delete, priority levels
Notes	Quick note-taking	Create, edit, timestamp tracking
Reminders	Time-based alerts	Date/time scheduling
Stats	Usage analytics	Task completion rates, productivity metrics

System Workflow



Technology Stack



Backend Architecture

Layer	Technology	Purpose
Web Framework	Flask 3.0	REST API, routing, session management
AI Engine	Ollama	Local LLM inference
Database	MongoDB	Document storage, chat history
Search	Multi-provider	Google API, DuckDuckGo, web scraping
TTS	ElevenLabs API	Voice synthesis
Code Execution	Piston API	Sandboxed code running

Frontend Technologies

Technology	Purpose
HTML5/CSS3	Structure and styling
JavaScript (ES6+)	Interactive functionality
Chart.js	Data visualization
Mermaid.js	Diagram rendering
Highlight.js	Code syntax highlighting
Marked.js	Markdown parsing

AI Models (Ollama)

Model	Alias	Parameters	Use Case
gpt-oss:20b-cloud	Laplacian Core	20B	General purpose, complex tasks
phi3:mini	Laplacian Lite	3.8B	Fast responses, lightweight tasks
qwen3-coder:480b-cloud	Laplacian Coder	Variable	Code generation, debugging

Dependencies

```
Flask==3.0.0           # Web framework
requests==2.31.0       # HTTP client
python-dotenv==1.0.0   # Environment management
beautifulsoup4==4.12.2 # Web scraping
googlesearch-python==1.2.4 # Search integration
PyPDF2==3.0.1          # PDF processing
python-docx==1.1.0     # DOCX processing
openpyxl==3.1.2        # Excel processing
pymongo==4.6.1         # MongoDB driver
Werkzeug==3.0.1        # WSGI utilities
pydub==0.25.1          # Audio processing
SpeechRecognition==3.10.0 # Voice input
```

Project Structure

```
laplacian-ai/
├─ app/                                # Application package
│   ├── __init__.py                    # Application factory
│   ├── config/                        # Configuration
│   │   ├── settings.py                # Environment settings
│   │   └─ constants.py                # Static constants
│   ├── routes/                        # API endpoints
│   │   ├── main.py                    # Index route
│   │   ├── chat.py                    # Chat API
│   │   ├── models.py                  # Model selection
│   │   ├── search.py                  # Web search
│   │   ├── code.py                    # Code execution
│   │   ├── dociq.py                   # Document intelligence
│   │   ├── viziq.py                   # Data visualization
│   │   └─ productivity.py              # Tasks, notes, reminders
│   ├── services/                      # Business logic
│   │   ├── ai_service.py              # AI response generation
│   │   ├── chat_service.py            # Conversation management
│   │   ├── search_service.py           # Web search providers
│   │   ├── speech_service.py           # Text-to-speech
│   │   ├── code_service.py             # Code execution
│   │   ├── dociq_service.py            # Document RAG
│   │   ├── viziq_service.py           # Data analytics
│   │   └─ productivity_service.py
│   └─ utils/                          # Utilities
│       ├── session.py                 # Session management
│       ├── file_helpers.py             # File operations
│       ├── text_processing.py          # Document parsing
│       └─ data_processing.py           # Data analysis
├─ templates/                          # HTML templates
│   └─ index.html                      # Main SPA template
├─ static/                             # Static assets
│   ├── style.css                      # Main styles
│   ├── enhancements.css               # UI enhancements
│   ├── script.js                      # Frontend logic
│   └─ logo.png                       # Brand assets
├─ uploads/                            # File upload directory
├─ database.py                         # MongoDB integration
└─ run.py                             # Application entry point
```

└─ requirements.txt	# Python dependencies
└─ .env	# Environment variables

Installation

Prerequisites

- Python 3.10 or higher
- Ollama (for local LLM)
- MongoDB (optional, falls back to in-memory)
- Node.js (for frontend development)

Quick Start

```
# Clone the repository
git clone https://github.com/perfionix-ai/laplacian-ai.git
cd laplacian-ai

# Create virtual environment
python -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Pull Ollama models
ollama pull phi3:mini
ollama pull qwen3-coder

# Configure environment
cp .env.example .env
# Edit .env with your settings

# Run the application
python run.py
```

Environment Configuration

```
# Flask Configuration
FLASK_SECRET_KEY=your-secure-secret-key
FLASK_DEBUG=True
FLASK_ENV=development

# Ollama Configuration
GPT_SERVER_URL=http://localhost:11434/api/chat
GPT_MODEL=gpt-oss:20b-cloud
LITE_MODEL=phi3:mini
CODER_MODEL=qwen3-coder:480b-cloud

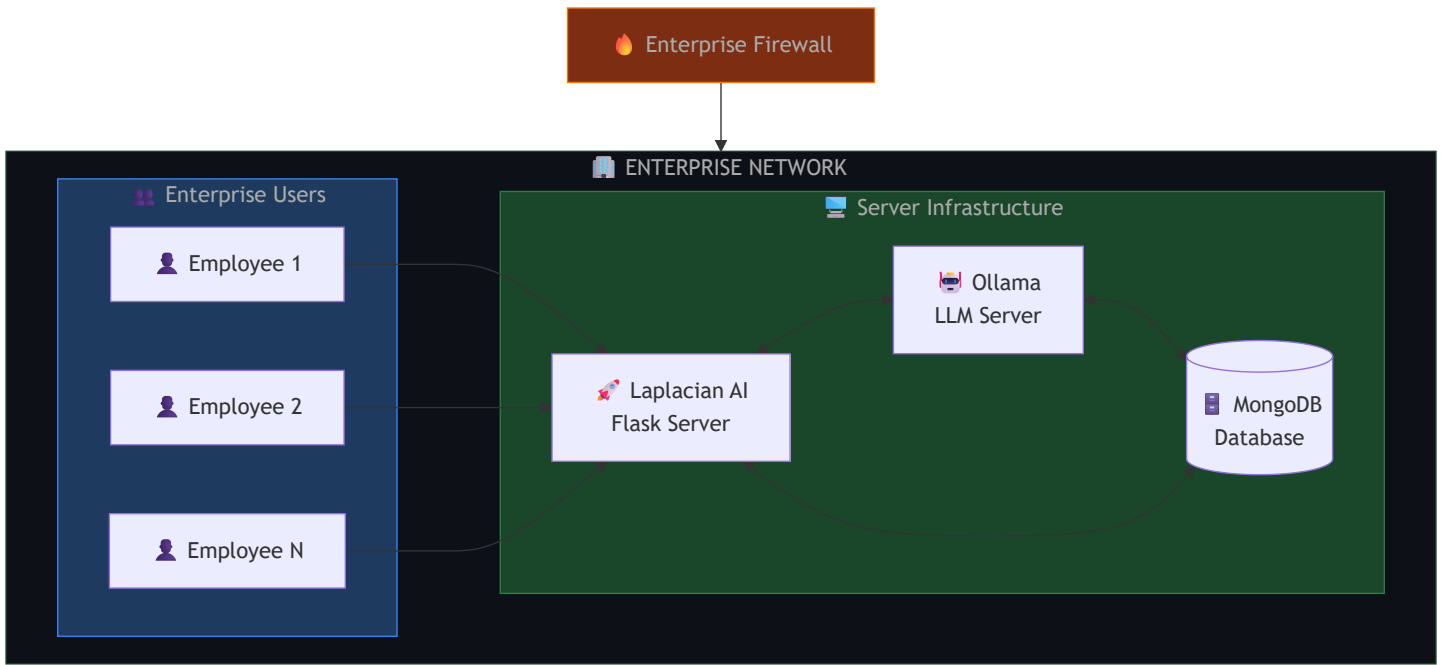
# MongoDB (optional)
MONGODB_URI=mongodb://localhost:27017
MONGODB_DB=laplacian_ai

# Optional APIs
ELEVENLABS_API_KEY=your-elevenlabs-key
GOOGLE_API_KEY=your-google-api-key
GOOGLE_CSE_ID=your-google-cse-id
```

Deployment Options

On-Premise Deployment

Recommended for enterprises requiring maximum data security:



Indian Cloud Deployment

For organizations preferring managed infrastructure:

- **AWS Mumbai Region** (ap-south-1)
- **Azure Central India**
- **Google Cloud Mumbai**
- **Indian Cloud Providers** (Yotta, CtrlS, NxtGen)

Security & Compliance

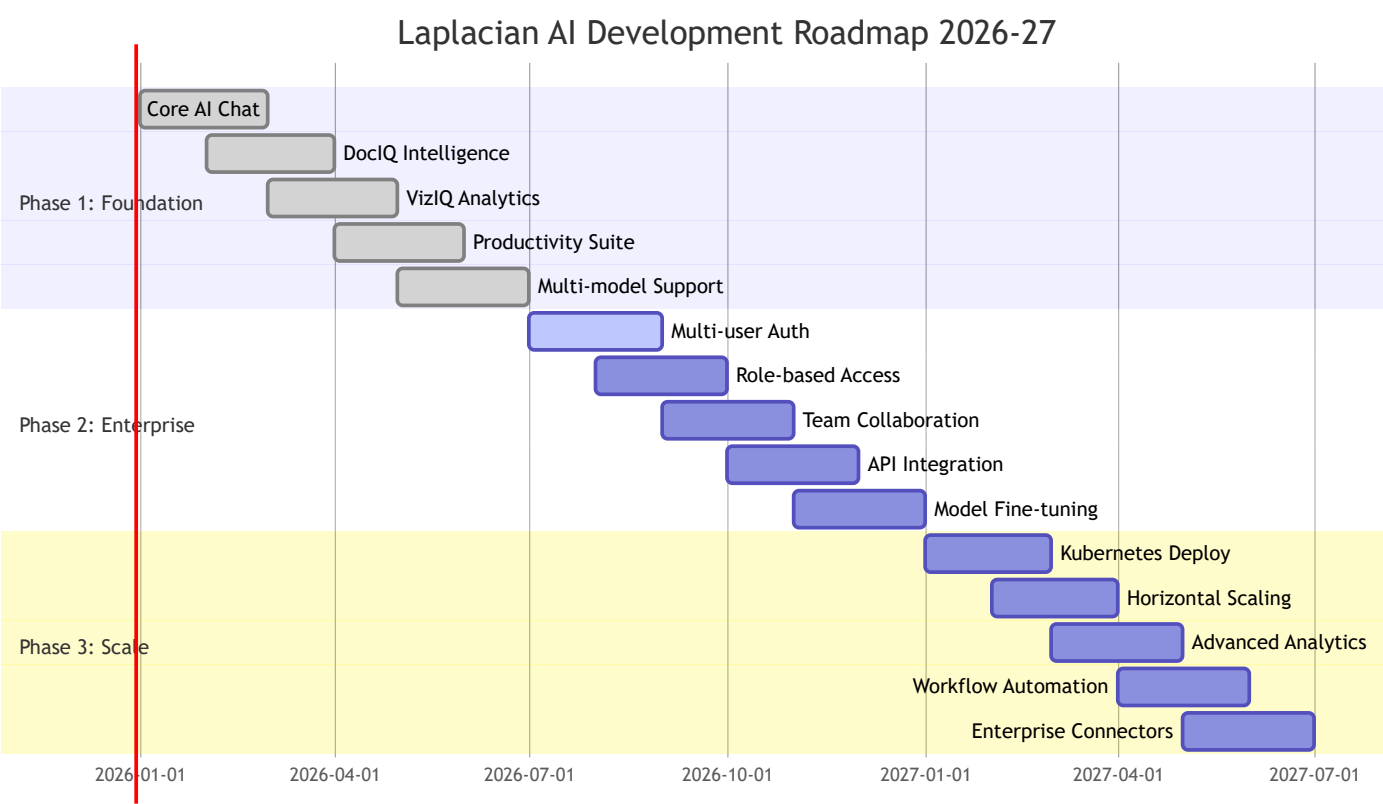
Data Protection Measures

Measure	Implementation
Data Localization	All processing within Indian servers
Encryption	TLS 1.3 for transit, AES-256 for storage
Access Control	Session-based authentication
Audit Logging	Comprehensive activity logs
Data Isolation	Per-session document separation

Regulatory Compliance

- ☒ **DPDP Act 2023** - Full compliance with data localization
- ☒ **IT Act 2000** - Adherence to intermediary guidelines
- ☒ **ISO 27001** - Security management standards
- ☒ **SOC 2 Type II** - Enterprise security controls

Roadmap



Phase 1: Foundation ☒ (Current)

- ☒ Core AI Chat functionality
- ☒ DocIQ document intelligence
- ☒ VizIQ data visualization
- ☒ Productivity suite
- ☒ Multi-model support

Phase 2: Enterprise 🚧 (Q3-Q4 2026)

- ☐ Multi-user authentication
- ☐ Role-based access control
- ☐ Team collaboration features
- ☐ API access for integration
- ☐ Custom model fine-tuning

Phase 3: Scale 📋 (Q1-Q2 2027)

- ☐ Kubernetes deployment
- ☐ Horizontal scaling
- ☐ Advanced analytics
- ☐ Workflow automation
- ☐ Enterprise connectors

Contributing

We welcome contributions from the developer community. Please read our [Contributing Guidelines](#) before submitting pull requests.

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"Securing India's AI Future, One Enterprise at a Time"